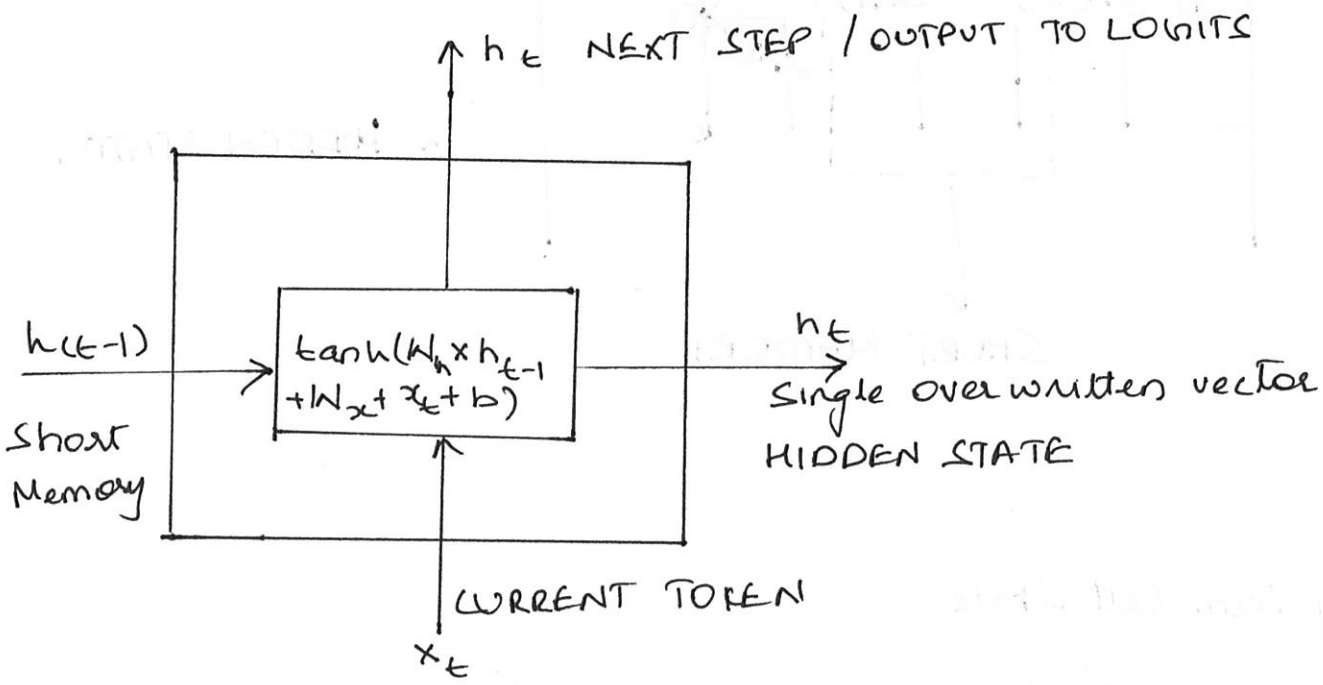
ASSIGNMENT – LONG SHORT TERM MEMORY

SUBMITTED - BY - DAMODHAR, VAMSHI

from Vanilla RNN Bottlenecks to LSTM Solutions

RNN CELL



BOTTLENECKS

① Memory Pressure

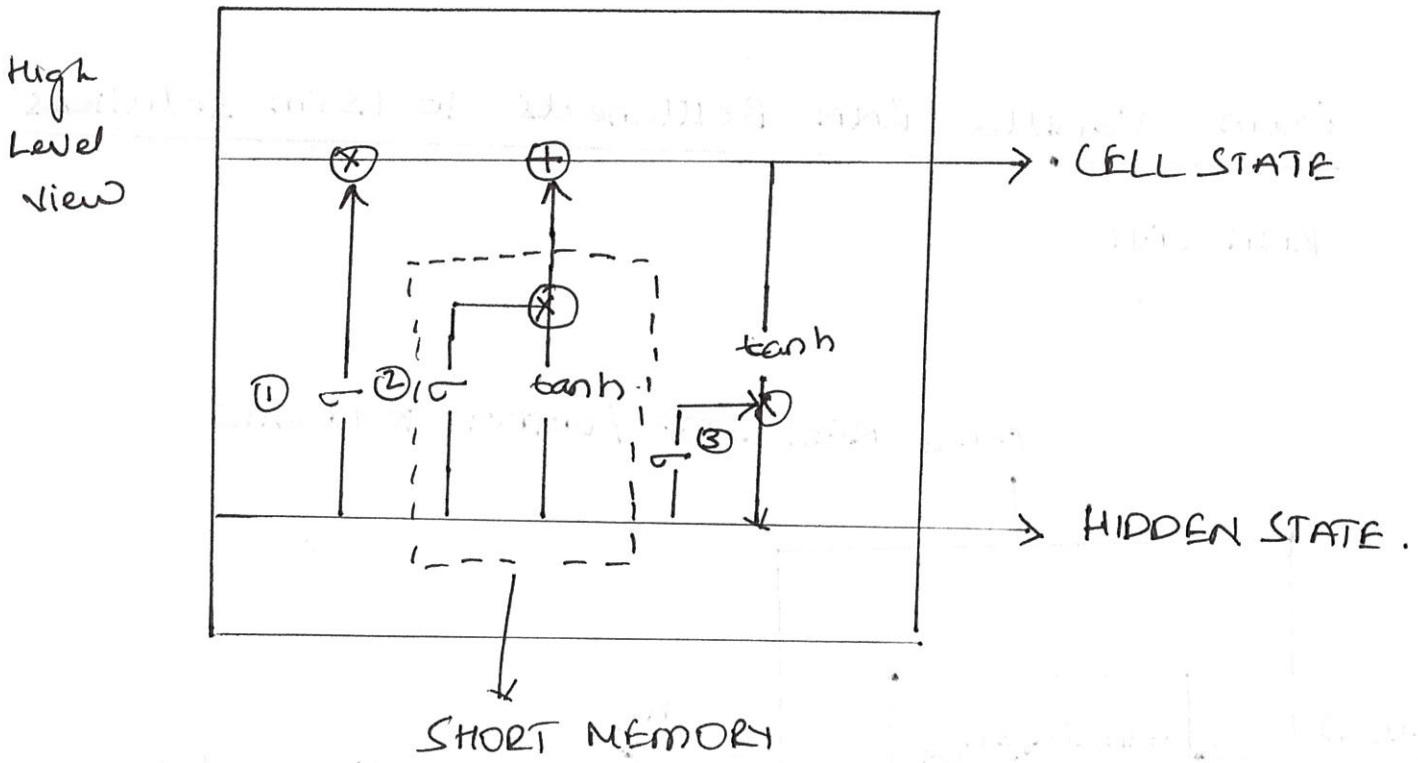
② Vanishing Gradient Problem.

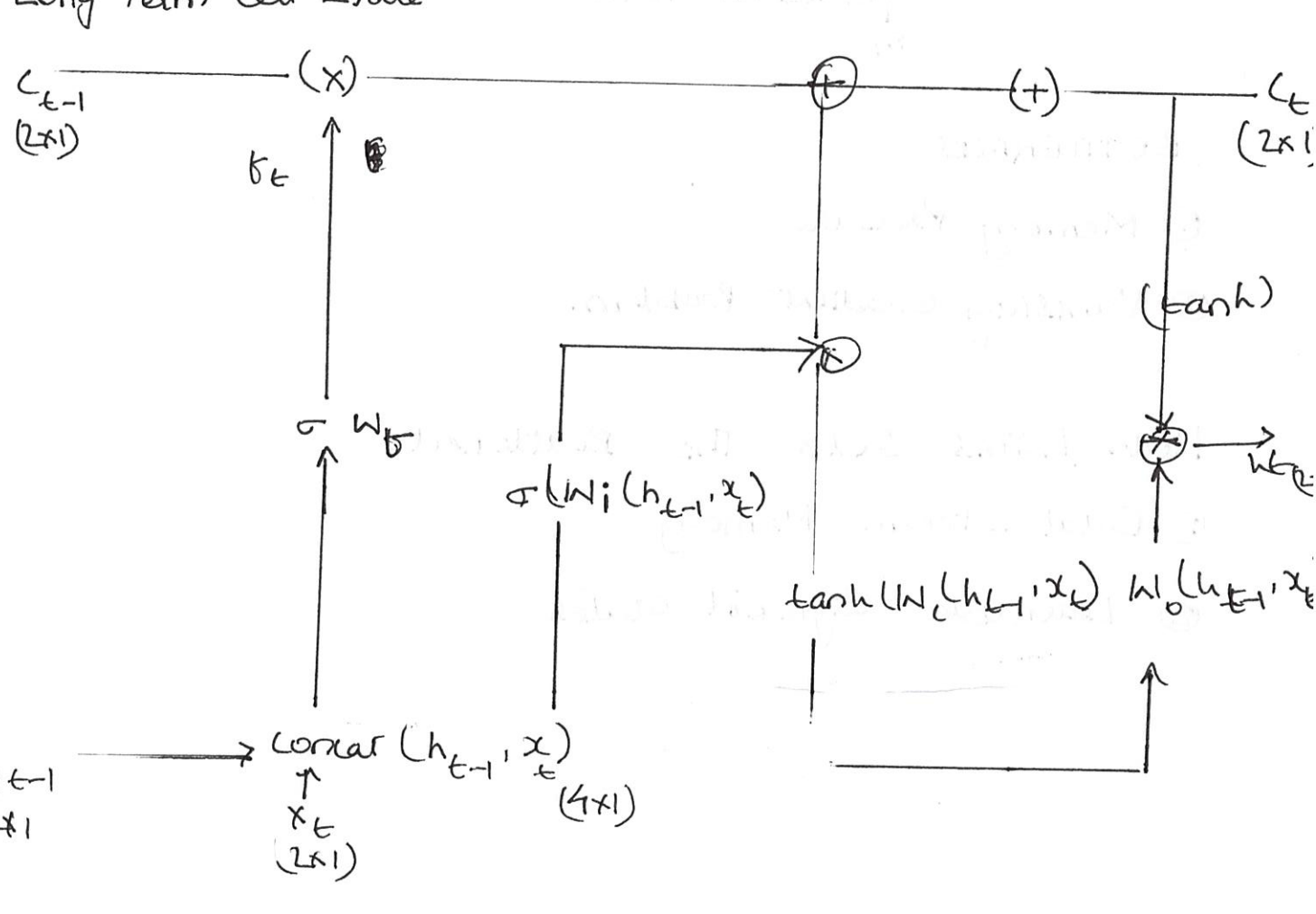
HOW LSTMS Solve This Bottlenecks

① Dual Stream Memory

② Learnable Sigmoid Gates

LSTM CELL INTERNALS





6 Governing Equations

① Forget Gate (How much of C_{t-1} to keep)

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

② Input Gate (How much of new candidate info to write)

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

③ Output Gate (How much of updated cell memory to expose as h_t)

$$O_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$$

④ Candidate Cell State (New proposed semantic content)

$$\bar{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)$$

⑤ Cell State Update

$$C_t = f_t \odot C_{t-1} + i_t \odot \bar{C}_t$$

⑥ Hidden State Update

$$h_t = O_t \odot \tanh(C_t)$$

Vocabulary Size: $V = 5$

Hidden State size: $h = 2$

Token Embedding size: $e = 2$

Context size: $c = 3$

Concatenated Input dimension: $(h + e) = (2 + 2) = 4$

Vocabulary Mapping ($V = 5$)

Token ID	Word	Semantic Role
0	"I"	Subject Pronoun
1	"love"	Action / sentiment verb
2	"deep"	Technical Modifier
3	"learning"	Target Technical noun
4	"<EOS>"	End-of-Sentence Token

Input Context Sequence ($c = 3$ tokens):

Context Words = ["I", "love", "deep"] => Context Token IDs = [0, 1, 2]

Step 1:

$$E = \begin{bmatrix} 1.0 & 0.0 \\ 0.5 & 1.0 \\ 0.0 & 1.0 \\ -0.5 & 1.0 \\ 0.0 & -1.0 \end{bmatrix}_{(5\times2)} \begin{matrix} \leftarrow ID_0 \text{ ("I")} \\ \leftarrow ID_1 \text{ ("love")} \\ \leftarrow ID_2 \text{ ("deep")} \\ \leftarrow ID_3 \text{ ("learning")} \\ \leftarrow ID_4 \text{ (" ")} \end{matrix}$$

Columns represent dimensions

column 1 → Subject/syntax

column 2 → Sentiment/Technical Topic

$$x_1 = E^T_{(2\times5)} \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}_{(5\times1)} = \begin{bmatrix} 1.0 \\ 0.0 \end{bmatrix}_{(2\times1)}$$

$$x_2 = E^T_{(2\times5)} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}_{(5\times1)} = \begin{bmatrix} 0.5 \\ 1.0 \end{bmatrix}_{(2\times1)}$$

$$x_3 = E^T_{(2\times5)} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}_{(5\times1)} = \begin{bmatrix} 0.0 \\ 1.0 \end{bmatrix}_{(2\times1)}$$

LSTM & logits Layer Parameters (With Explicit shapes)

Initial states ($t=0$)

$$h_0 = \begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix}_{(2\times1)}, \quad C_0 = \begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix}_{(2\times1)}$$

LSTM Gate Weights $(h \times (h+e)) = (2\times4)$ and biases $(h\times1) = (2\times1)$

1) Forget Gate Parameters

$$W_f = \begin{bmatrix} 0.5 & 0.0 & 0.5 & 0.2 \\ 0.0 & 0.5 & -0.2 & 0.5 \end{bmatrix}_{(2\times4)}, \quad b_f = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}_{(2\times1)}$$

2) Input Gate Parameters

$$W_i = \begin{bmatrix} 0.4 & -0.2 & 0.8 & 0.2 \\ -0.2 & 0.4 & 0.2 & 0.8 \end{bmatrix}_{(2\times4)}, \quad b_i = \begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix}_{(2\times1)}$$

3) Output Gate Parameters

$$W_o = \begin{bmatrix} 0.5 & 0.2 & 0.6 & 0.2 \\ 0.2 & 0.5 & 0.2 & 0.6 \end{bmatrix}_{(2\times4)}, \quad b_o = \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix}_{(2\times1)}$$

4) Candidate Cell Parameters

$$W_c = \begin{bmatrix} 0.5 & 0.0 & 1.0 & -0.5 \\ 0.0 & 0.5 & -0.5 & 1.0 \end{bmatrix}_{(2\times4)}, \quad b_c = \begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix}_{(2\times1)}$$

5) Dense Logits Layer Parameters $(V\times h) = (5\times2)$ and $(V\times1) = (5\times1)$

$$W_{out} = \begin{bmatrix} 0.5 & -1.0 \\ 0.8 & 0.2 \\ -0.4 & 0.6 \\ -0.5 & 2.0 \\ -0.2 & -0.8 \end{bmatrix}_{(5\times2)}, \quad b_{out} = \begin{bmatrix} 0.0 \\ 0.0 \\ 0.1 \\ 0.5 \\ -0.2 \end{bmatrix}_{(5\times1)}$$

LSTM Forward Pass at Time Step t=1 (Token 0: "I")

A) Concatenate previous hidden state (h_0) and current Embedding (x_1)

$$[h_0, x_1] = \begin{bmatrix} h_{0,1} \\ h_{0,2} \\ x_{1,1} \\ x_{1,2} \end{bmatrix}_{(4\times1)} = \begin{bmatrix} 0.0 \\ 0.0 \\ 1.0 \\ 0.0 \end{bmatrix}_{(4\times1)}$$

B) Forget Gate $f_1 = \sigma(W_f[h_0, x_1] + b_f)$

$$W_f[h_0, x_1] + b_f = \begin{bmatrix} 0.5 & 0.0 & 0.5 & 0.2 \\ 0.0 & 0.5 & -0.2 & 0.5 \end{bmatrix}_{(2\times4)} \begin{bmatrix} 0.0 \\ 0.0 \\ 1.0 \\ 0.0 \end{bmatrix}_{(4\times1)} + \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}_{(2\times1)} = \begin{bmatrix} 1.0 \\ 0.3 \end{bmatrix}_{(2\times1)}$$

$$f_1 = \sigma\left(\begin{bmatrix} 1.0 \\ 0.3 \end{bmatrix}\right) = \begin{bmatrix} 0.7311 \\ 0.5744 \end{bmatrix}$$

C) Input Gate $i_1 = \sigma(W_i[h_0, x_1] + b_i)$

$$W_i[h_0, x_1] + b_i = \begin{bmatrix} 0.4 & -0.2 & 0.8 & 0.2 \\ -0.2 & 0.4 & 0.2 & 0.8 \end{bmatrix}_{(2\times4)} \begin{bmatrix} 0.0 \\ 0.0 \\ 1.0 \\ 0.0 \end{bmatrix}_{(4\times1)} + \begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix}_{(2\times1)} = \begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}_{(2\times1)}$$

$$i_1 = \sigma\left(\begin{bmatrix} 0.8 \\ 0.2 \end{bmatrix}\right) = \begin{bmatrix} 0.69 \\ 0.54 \end{bmatrix}_{(2\times1)}$$

D) Output Gate $O_1 = \sigma(W_o[h_0, x_1] + b_o)$

$$W_o[h_0, x_1] + b_o = \begin{bmatrix} 0.5 & 0.2 & 0.6 & 0.2 \\ 0.2 & 0.5 & 0.2 & 0.6 \end{bmatrix}_{(2\times4)} \begin{bmatrix} 0.0 \\ 0.0 \\ 1.0 \\ 0.0 \end{bmatrix}_{(4\times1)} + \begin{bmatrix} 0.0 \\ 0.1 \end{bmatrix}_{(2\times1)} = \begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix}_{(2\times1)}$$

$$O_1 = \sigma\left(\begin{bmatrix} 0.7 \\ 0.3 \end{bmatrix}\right) = \begin{bmatrix} 0.66 \\ 0.57 \end{bmatrix}_{(2\times1)}$$

E) Candidate cell state $\bar{C}_1 = \tanh(W_c[h_0, x_1] + b_c)$

$$W_c[h_0, x_1] + b_c = \begin{bmatrix} 0.5 & 0.0 & 1.0 & -0.5 \\ 0.0 & 0.5 & -0.5 & 1.0 \end{bmatrix}_{(2\times4)} \begin{bmatrix} 0.0 \\ 0.0 \\ 1.0 \\ 0.0 \end{bmatrix}_{(4\times1)} + \begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix}_{(2\times1)} = \begin{bmatrix} 1.0 \\ -0.5 \end{bmatrix}_{(2\times1)}$$

$$\bar{C}_1 = \tanh\left(\begin{bmatrix} 1.0 \\ -0.5 \end{bmatrix}\right) = \begin{bmatrix} 0.76 \\ -0.46 \end{bmatrix}_{(2\times1)}$$

F) Cell state C_1 & Hidden state h_1 Updates

$$C_1 = f_1 \odot C_0 + i_1 \odot \bar{C}_1 = \begin{bmatrix} 0.73 \\ 0.57 \end{bmatrix}_{(2\times1)} \odot \begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix}_{(2\times1)} + \begin{bmatrix} 0.69 \\ 0.54 \end{bmatrix}_{(2\times1)} \odot \begin{bmatrix} 0.76 \\ -0.96 \end{bmatrix}_{(2\times1)} = \begin{bmatrix} 0.52 \\ -0.25 \end{bmatrix}_{(2\times1)}$$

$$h_1 = O_1 \odot \tanh(C_1) = \begin{bmatrix} 0.66 \\ 0.57 \end{bmatrix}_{(2\times1)} \odot \tanh\left(\begin{bmatrix} 0.52 \\ -0.25 \end{bmatrix}\right) = \begin{bmatrix} 0.32 \\ -0.14 \end{bmatrix}_{(2\times1)}$$

LSTM Forward pass at Time step $t=2$ (Token 1 : "love")

A) Concatenate Previous hidden state h_1 and current Embedding x_2

$$[h_1, x_2] = \begin{bmatrix} 0.32 \\ -0.14 \\ 0.5 \\ 1.0 \end{bmatrix}_{(4\times1)}$$

B) Forget Gate

$$f_2 = \sigma(W_f [h_1, x_2] + b_f) = \sigma\left(\begin{bmatrix} 0.5 & 0.0 & 0.5 & 0.2 \\ 0.0 & 0.5 & -0.2 & 0.5 \end{bmatrix}_{(2\times4)} \begin{bmatrix} 0.32 \\ -0.14 \\ 0.50 \\ 1.00 \end{bmatrix}_{(4\times1)} + \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}\right) = \begin{bmatrix} 0.75 \\ 0.69 \end{bmatrix}_{(2\times1)}$$

C) Input Gate ($i_2 = \sigma(W_i [h_1, x_2] + b_i)$)

$$i_2 = \sigma\left(\begin{bmatrix} 0.4 & -0.2 & 0.8 & 0.2 \\ -0.2 & 0.4 & 0.2 & 0.8 \end{bmatrix}_{(2\times4)} \begin{bmatrix} 0.33 \\ -0.14 \\ 0.50 \\ 1.00 \end{bmatrix}_{(4\times1)} + \begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix}_{(2\times1)}\right)$$

$$= \begin{bmatrix} 0.68 \\ 0.68 \end{bmatrix}_{(2\times1)}$$

D) Output Gate $O_2 = \sigma(W_o[h_1, x_2] + b_o)$

$$O_2 = \sigma\left(\begin{bmatrix} 0.5 & 0.2 & 0.6 & 0.2 \\ 0.2 & 0.5 & 0.2 & 0.6 \end{bmatrix}_{(2\times4)} \begin{bmatrix} 0.32 \\ -0.14 \\ 0.50 \\ 1.00 \end{bmatrix}_{(4\times1)} + \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix}_{(2\times1)}\right)$$

$$O_2 = \sigma\left(\begin{bmatrix} 0.73 \\ 0.79 \end{bmatrix}\right) = \begin{bmatrix} 0.67 \\ 0.68 \end{bmatrix}_{(2\times1)}$$

E) Candidate Cell state $\bar{C}_2 = \tanh(W_c[h_1, x_2] + b_c)$

$$\bar{C}_2 = \tanh\left(\begin{bmatrix} 0.5 & 0.0 & 1.0 & -0.5 \\ 0.0 & 0.5 & -0.5 & 1.0 \end{bmatrix}_{(2\times4)} \begin{bmatrix} 0.32 \\ -0.14 \\ 0.50 \\ 1.00 \end{bmatrix}_{(4\times1)} + \begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix}_{(2\times1)}\right)$$

$$\bar{C}_2 = \tanh\left(\begin{bmatrix} 0.16 \\ 0.67 \end{bmatrix}\right) = \begin{bmatrix} 0.15 \\ 0.59 \end{bmatrix}_{(2\times1)}$$

F) Cell state (C2) & Hidden State (h_2) Updates

$$C_2 = f_2 \odot C_1 + i_2 \odot \bar{C}_2$$

$$C_2 = \begin{bmatrix} 0.75 \\ 0.69 \end{bmatrix}_{(2\times1)} \odot \begin{bmatrix} 0.52 \\ -0.25 \end{bmatrix}_{(2\times1)} + \begin{bmatrix} 0.68 \\ 0.68 \end{bmatrix}_{(2\times1)} \odot \begin{bmatrix} 0.15 \\ 0.59 \end{bmatrix}_{(2\times1)}$$

$$C_2 = \begin{bmatrix} 0.50 \\ 0.22 \end{bmatrix}_{(2\times1)}$$

$$h_2 = O_2 \odot \tanh(C_2) = \begin{bmatrix} 0.67 \\ 0.68 \end{bmatrix}_{(2\times1)} \odot \tanh\left(\begin{bmatrix} 0.67 \\ 0.68 \end{bmatrix}\right)$$

$$h = \begin{bmatrix} 0.31 \\ 0.15 \end{bmatrix}_{(2\times1)}$$

LSTM forward pass at Time Step t=3 (Token 2 : "deep")

→ Concat previous hidden state h_2 and current embeddings

$$[h_2, x_3] = \begin{bmatrix} 0.3142 \\ 0.1542 \\ 0.0000 \\ 1.0000 \end{bmatrix}_{(4\times1)}$$

→ Forget Gate $f_3 = \sigma(W_f(h_2, x_3) + b_f)$

$$\begin{bmatrix} 0.5 & 0.0 & 0.5 & 0.2 \\ 0 & 0.5 & -0.2 & 0.5 \end{bmatrix}_{(2\times4)} \times \begin{bmatrix} 0.3142 \\ 0.1542 \\ 0.00 \\ 1.00 \end{bmatrix}_{(4\times1)} + \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}_{(2\times1)}$$

$$= \begin{bmatrix} 0.3571 \\ 0.5771 \end{bmatrix}_{(2\times1)} + \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}_{2\times1} = \begin{bmatrix} 0.8571 \\ 1.0771 \end{bmatrix}_{(2\times1)}$$

→ Input Gate $i_3 = \sigma[W_i[h_2, x_3] + b_i]$

$$= \begin{bmatrix} 0.4 & -0.2 & 0.8 & 0.2 \\ -0.2 & 0.4 & 0.2 & 0.8 \end{bmatrix}_{(2\times4)} \begin{bmatrix} 0.3142 \\ 0.1542 \\ 0.00 \\ 1.00 \end{bmatrix}_{(4\times1)} + \begin{pmatrix} 0.0 \\ 0.6 \end{pmatrix}_{2\times1}$$

$$= \begin{bmatrix} 0.2948 \\ 0.7988 \end{bmatrix}_{(2\times1)} \quad i_3 = \sigma \begin{bmatrix} 0.2948 \\ 0.7988 \end{bmatrix} = \begin{bmatrix} 0.5732 \\ 0.6897 \end{bmatrix}_{2\times1}$$

→ Output Gate $O_3 = \sigma(W_o[h_2, x_3] + b_o)$

$$= \begin{bmatrix} 0.5 & 0.2 & 0.6 & 0.2 \\ 0.2 & 0.5 & 0.2 & 0.6 \end{bmatrix}_{(2\times4)} \begin{bmatrix} 0.3142 \\ 0.1542 \\ 0.0 \\ 1.00 \end{bmatrix}_{4\times1} + \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix}_{(2\times1)}$$

$$= \quad \cancel{0.1571} \quad \cancel{0.03081}$$

$$\begin{bmatrix} 0.3879 \\ 0.7399 \end{bmatrix}_{(2\times1)} + \begin{bmatrix} 0.1 \\ 0.1 \end{bmatrix}_{(2\times1)} = \begin{bmatrix} 0.4879 \\ 0.8399 \end{bmatrix}_{(2\times1)}$$

$$O_3 = \sigma\begin{pmatrix} 0.4879 \\ 0.8399 \end{pmatrix} = \begin{bmatrix} 0.6196 \\ 0.6984 \end{bmatrix}_{(2\times1)}$$

→ Candidate Cell State $C_3 = \tanh(W_c[h_2, x_3] + b_c)$

$$= \tanh\underset{(2\times4)}{\begin{bmatrix} 0.5 & 0.0 & 1.0 & -0.5 \\ 0 & 0.5 & -0.5 & 1.0 \end{bmatrix}} \underset{4\times1}{\begin{bmatrix} 0.3142 \\ 0.1542 \\ 0.0 \\ 1.00 \end{bmatrix}} + \begin{bmatrix} 0.0 \\ 0.0 \end{bmatrix}_{(2\times1)}$$

$$= \tanh\begin{bmatrix} -0.3429 \\ 1.0771 \end{bmatrix}_{(2\times1)} = \begin{bmatrix} -0.3301 \\ 0.7921 \end{bmatrix}_{(2\times1)}$$

final Context Cell State C_3 & Final Hidden State h_3 (T=3)

$$C_3 = f_3 \odot C_2 + i_3 \odot \tilde{C}_3 = \underset{(2\times1)}{\begin{bmatrix} 0.7021 \\ 0.7459 \end{bmatrix}} \cdot \underset{(2\times1)}{\begin{bmatrix} 0.5040 \\ 0.2279 \end{bmatrix}} +$$

$$\underset{(2\times1)}{\begin{bmatrix} 0.5732 \\ 0.6097 \end{bmatrix}} \odot \begin{bmatrix} -0.3301 \\ 0.7921 \end{bmatrix}_{(2\times1)}$$

$$= \begin{bmatrix} 0.1647 \\ 0.7162 \end{bmatrix}_{(2\times1)}$$

$$h_3 = O_3 \odot \tanh(C_3) = \begin{bmatrix} 0.6196 \\ 0.6984 \end{bmatrix} \odot \tanh\left(\begin{bmatrix} 0.1647 \\ 0.7162 \end{bmatrix}\right)$$

$$= \begin{bmatrix} 0.1011 \\ 0.4292 \end{bmatrix}_{(2\times1)}$$

DENSE LOGITS LAYER

Eq: $y = (W_{out} h_3 + b_{out})$

Using hidden state $h_3 = \begin{bmatrix} 0.1011 \\ 0.4292 \end{bmatrix}$ (2×1)

$$\begin{bmatrix} 0.5 & -1.0 \\ 0.8 & 0.2 \\ -0.4 & 0.6 \\ -0.5 & 2.0 \\ -0.2 & -0.8 \end{bmatrix}_{(5\times2)} \times \begin{bmatrix} 0.1011 \\ 0.4292 \end{bmatrix}_{(2\times1)} + \begin{bmatrix} 0.0 \\ 0.0 \\ 0.1 \\ 0.5 \\ -0.2 \end{bmatrix}_{5\times1}$$

$$y = \begin{bmatrix} -0.3787 \\ 0.1667 \\ 0.3171 \\ 1.3079 \\ -0.5636 \end{bmatrix}_{(5\times1)}$$

SOFTMAX Probability Distribution & Token Selection.

$$\text{Softmax}(y) = \begin{bmatrix} 0.0912 \\ 0.1574 \\ 0.1929 \\ 0.4927 \\ 0.0758 \end{bmatrix}_{(5\times1)} \longrightarrow \text{WINNER.}$$

Predicted Next word = "learning"